



Northeastern University
College of Engineering

Control Systems Engineering (ME5659)

Project Report

Design and Control of a Single-Axis Magnetic Levitation

System: Stability Analysis and Implementation of State

Space Control Strategies

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1- Introduction

Magnetic levitation (maglev) systems are revolutionizing industries like transportation, precision manufacturing, and biomedical research. Maglev trains, such as Japan's SCMaglev, offer high-speed travel over 600 km/h with lower energy consumption, minimal environmental impact, and reduced maintenance compared to conventional rail [1]. These systems are a promising alternative for sustainable urban transportation, alleviating congestion and reducing carbon emissions [2].

Maglev technology is also transforming precision engineering, providing highly accurate, vibration-free motion control for applications such as semiconductor manufacturing and high-resolution imaging. The growing need for robust control strategies to maintain stability and precision in these systems is driving research in the field [3], [4], [5]. Maglev technology eliminates friction, leading to energy savings and enhanced mechanical reliability. Its applications in high-speed transportation and precision engineering make it essential for sustainable development. Maglev trains offer an eco-friendly alternative to conventional rail and air travel, especially in congested urban areas [2], [6]. Additionally, maglev-based nano-positioning systems are vital for precision tasks in fields like optical systems and biomedical research [7]. In addition, the scientific challenge of stabilizing nonlinear maglev systems advances control theory, impacting a wide range of fields, including robotics, aerospace, and energy systems [8], [9]. Control of maglev systems has evolved from simple PID controllers to more advanced methods like state-space control, sliding mode control, and backstepping. While classical methods are effective for small perturbations, they struggle with the system's inherent nonlinearities [10]. Nonlinear methods, such as sliding mode control, offer robustness but can introduce chattering, while backstepping enhances stability by managing nonlinearities [11], [12]. Adaptive and machine learning techniques are being explored to adjust control parameters dynamically in real-time.

This project aims to implement output and state feedback controllers, with and without observers for control position purposes of a single-axis maglev system, where a steel ball is levitated by electromagnetic force. The system's nonlinear dynamics and requires some considerations in implementing an appropriate control policies. We will develop a mathematical model of the system using Newton's laws and electromagnetic equations, simplifying it through linearization. PID controllers will be implemented initially, followed by state-space techniques like pole placement in presence of an observer. The performance of these controllers will be tested using MATLAB/Simulink.

2- Modeling of the system

A single-axis magnetic levitation system comprises an electromagnet and a steel ball. The upward electromagnetic force generated by the coil counteracts the downward gravitational force acting on the steel ball. The system exhibits nonlinear dynamics, as the electromagnetic force depends on the square of the coil current and inversely on the ball's displacement.

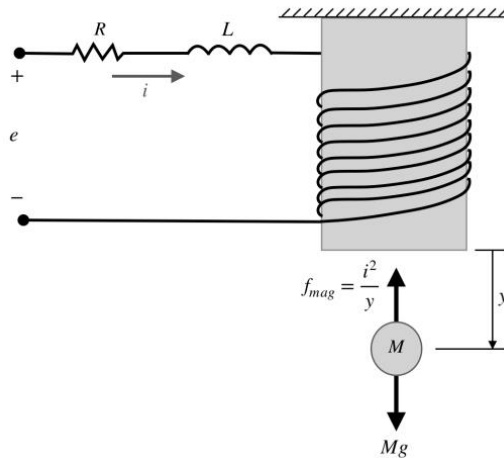


Figure 1 A basic example of a magnetic levitation system

Key parameters influencing the system's behavior include the applied voltage $e(t)$, the coil current $i(t)$, resistance R , inductance L , the steel ball's mass M , and its vertical position $y(t)$ as provided in Figure 1. These parameters collectively govern the interplay between electromagnetic and mechanical forces.

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Based on what described in Figure 1, the system's dynamics are derived from Kirchhoff's Voltage Law (KVL) for the electromagnetic circuit and Newton's Second Law for the forces acting on the steel ball. Using KVL and Newton's Second Law balances the electromagnetic force f_{mag} with gravity, the coil dynamics are described as:

$$\begin{aligned} M \left(\frac{d^2 y}{dt^2} \right) &= f_{mag} - Mg \\ f_{mag} &= \frac{i^2}{y} \end{aligned} \quad (1)$$

The nonlinear nature arises from the electromagnetic force expression, $f_{mag} = \frac{i^2}{y}$. By substituting f_{mag} into the force equation, the motion equation becomes:

$$M \left(\frac{d^2 y}{dt^2} \right) = \frac{i^2}{y} - Mg \quad (2)$$

Having the nonlinear representation of system dynamics in (2), we can derive the state space representation. To facilitate analysis and control design, the system's nonlinear dynamics are expressed in state-space form. The state variables are defined as $x_1 = y$ (position), $x_2 = y'$ (velocity), and $x_3 = i$ (current). The control input is the applied voltage $u(t) = e(t)$. Then, the state equations are:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= g - \frac{1}{M} \frac{x_3^2}{x_1} \\ \dot{x}_3 &= \frac{1}{L} (e - R x_3) \end{aligned} \quad (3)$$

One potential approach to implement classical control method on such a system is to linearize the system around a specific working condition, in that wise, we can simply approximate the dynamics with a linearized system and employ the appropriate control method.

To simplify control design, the nonlinear system is linearized around an equilibrium point where the ball is stationary. In this study, we assume that we aim to keep the ball in a fixed distance y_0 and define equilibrium point based on this fixed distance. At equilibrium, the electromagnetic force balances gravity, giving:

$$\dot{x}_1 = x_2 = \dot{x}_3 = 0 \rightarrow \dot{y}_0 = 0 \quad (4)$$

$$\dot{x}_2 = g - \frac{1}{M} \frac{x_3^2}{x_1} = f_2 = 0 \rightarrow i_0^2 = Mgy_0$$

$$\dot{x}_3 = \frac{1}{L}(e - Rx_3) = f_3 = 0 \rightarrow v_0 = Ri_0$$

The system can then be represented with parameters provided in (5).

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{g}{y_0} & 0 & -2\sqrt{\frac{g}{My_0}} \\ 0 & 0 & -\frac{R}{L} \end{bmatrix}; B = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \\ \frac{\partial f_3}{\partial u} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}; C = [1 \ 0 \ 0]; D = 0$$

$$X_0 = [0.05 \ 0 \ 0.22]$$
(5)

We ultimately can describe the evolution of states in linearized model with (5).

3- Stability analysis

In this project, we assume that we aim to keep the levitating mass at a defined distance. This requires providing a voltage command such that the position stabilizes at a constant value of 5 cm, and the velocity eventually approaches zero. However, considering the eigenvalues of the state matrix:

$$\det(sI - A) = \left(s^2 - \frac{g}{y_0}\right)\left(s + \frac{R}{L}\right) \rightarrow \lambda_{1,2} = \pm \sqrt{\frac{g}{y_0}}, \lambda_3 = -\frac{R}{L}$$
(6)

TABLE I Parameters of the system

Parameter	value
R	1 Ω
L	0.001 H
M	0.1 Kg
y_0	0.05 m

It is evident that, due to the presence of a right-half-plane pole $+\sqrt{\frac{g}{y_0}}$, the open-loop system is inherently unstable. We then need to design a state feedback controller to meet the desired performance specifications. For this problem, our goal is to design a controller with a maximum overshoot of 25% and a settling time of 2 seconds. In our design, we assume specific values for coil resistance, coil inductance, and ball mass, while maintaining the ball at a distance of 5 cm.

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Based on the specified parameters in TABLE I and the linearized representation of the system, then, the eigenvalues of such a system is:

$$\lambda_1 = 14.007, \lambda_2 = -14.007, \lambda_3 = -1000$$

We also employed Lyapunov stability analysis to show the stability status of the system. The Lyapunov function is selected as follows.

$$V(X) = x_1^2 + x_2^2 + x_3^2 \quad (7)$$

This function is positive definite, satisfying $V(X) > 0$ for $X \neq 0$ and $V(0) = 0$. The time derivative of $V(X)$, based on the system dynamics, is:

$$\dot{V}(X) = 2x_1x_2 + \left(g - \frac{x_3^2}{Mx_1^2}\right)^2 + \left(\frac{u - Rx_3}{L}\right)^2 - 117.17x_2x_3 - 2000x_3^2 \quad (8)$$

Using the system's parameters, it was found that $\dot{V}(x) = 0$, which indicates marginal stability under the given conditions. To determine asymptotic stability, $\dot{V}(x)$ must satisfy $\dot{V}(x) < 0$ for all $X \neq 0$. As part of this analysis, we tested the stability using the matrix equation $A^T P + PA = -Q$, where A is the state matrix, P is a symmetric positive definite matrix, and Q is a negative definite matrix.

$$A^T P + PA = -I$$

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} \quad (9)$$

Considering (5) and (9), we have:

$$p_{12} = \frac{-1}{392.4}, p_{21} = \frac{-391.4}{392.4}, p_{11} = -196.2p_{22} \rightarrow p_{11} = 1 \rightarrow p_{22} = \frac{-1}{196.2} \quad (10)$$

To check whether P is positive definite, we calculated the determinant of the leading principal minor which is approximately -1.0025 , which is negative. This indicates that the matrix P is not positive definite. Since the matrix P is not positive definite, the system fails to satisfy the Lyapunov stability criterion. Consequently, the magnetic levitation system is not asymptotically stable under the chosen conditions. Thus, we need to employ an appropriate control policy.

4- Control policies and controller design

In this section, two different control approaches are employed. First one is based on the output of the system and the second approach is based on the states of the system. First we consider a simple PID controller using output feedback. According to **Error! Reference source not found.**, we can derive the transfer function relation in frequency domain, between the input voltage and position of the levitating ball.

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D \rightarrow G(s) = \frac{88580}{s^3 + 10000s^2 - 196.2s - 1.962 \times 10^6} \quad (11)$$

For such a system, a PID controller with $K_p = 35, K_i = 25, \text{ and } K_d = 9$ have been employed to stabilize the linearized system.

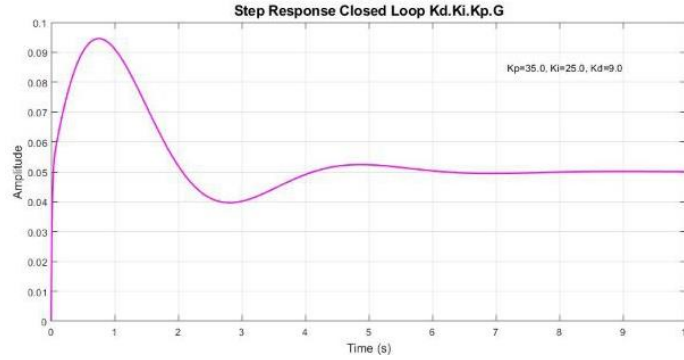


Figure 2 Tracking performance using PID controller with output feedback

To achieve enhanced control and improved system performance, it is essential to move beyond output feedback and explore more comprehensive approaches. State feedback control, which utilizes full state information, offers significant advantages in terms of stability, precision, and flexibility. By regulating all state variables, it ensures robust performance and enables advanced design techniques like pole placement to meet desired dynamic specifications. This makes state feedback control particularly well-suited for complex systems such as magnetic levitation. In this part, we aim to implement pole placement method for achieving desired performance. Thus, for the pole placement method, it is essential to verify the controllability of the states. To achieve this, we first construct the controllability matrix as follows.

$$Q = \begin{bmatrix} 0 & 0 & -8858.89 \\ 0 & -8858.89 & 8858893.84 \\ 100 & -100000 & 100000000 \end{bmatrix}$$

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The rank of matrix Q is 3, indicating that the controllability of this system is fully satisfied, as it is a full-rank matrix. With the states being controllable, we can arbitrarily place the controller's gain at desired locations using the pole placement approach. Based on the design specifications, the desired pole locations can be determined to achieve the specified maximum overshoot and settling time.

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.25 \rightarrow \zeta = 0.404$$

$$t_s = 2\text{ s} \rightarrow t_s = \frac{4}{\zeta\omega_n} \rightarrow \omega_n = 1.24\text{ rad/s}$$

$$\rightarrow s_{1,2} = -\zeta\omega_n + j\omega_n\sqrt{\zeta^2 - 1} \rightarrow s_{1,2} = -0.5 \pm j1.13$$
(12)

The characteristics equation of such a system is: $s^2 + s + 1.534 = 0$. But the original system's dimension is 3, so we place the third pole farther from the imaginary axis to minimize its effect on the system's transient response.

$$\rightarrow s_3 = -5$$

$$s^3 + 6s^2 + 6.534s + 7.67 = 0$$

Since the matrix is controllable, we can place the closed-loop system poles at the desired locations, resulting in the favorable response. To do so, we construct the state feedback control law as $u = -KX + K_g r$.

Additionally, a feedforward gain K_g is included to ensure steady-state tracking performance.

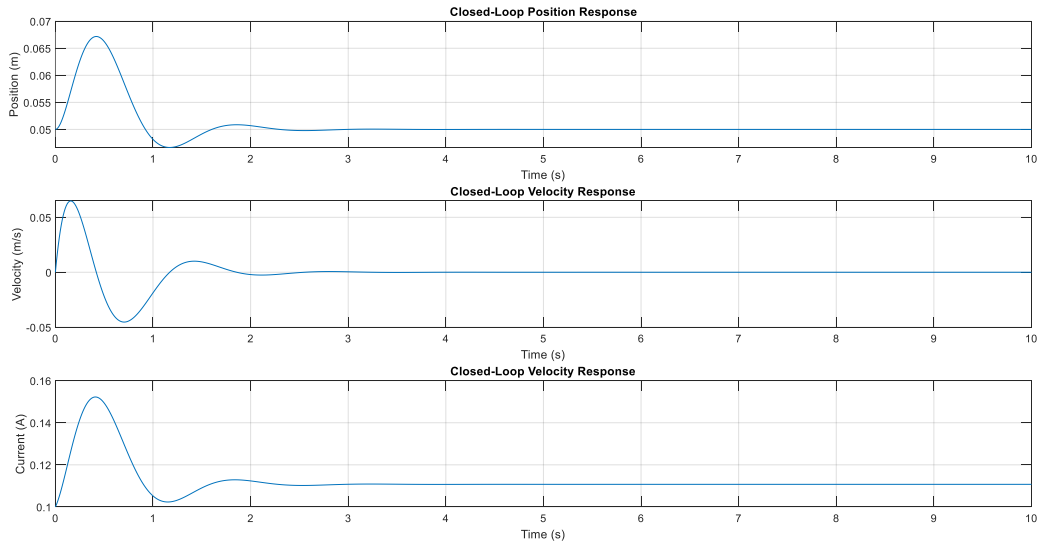


Figure 3 Position, velocity and current plots

$$\dot{x} = (A - BK)x + BK_g r, y = (C - DK)x + DK_g r \quad (13)$$

Using 'place' function in MATLAB we simply obtain the K matrix as follows:

$$K = [-0.2132 \quad -0.0272 \quad -9.9100], K_g = -0.0139 \quad (14)$$

After implementing this controller, the final result is shown in Figure 3.

Both velocity and position initial conditions were set to zero initially. After that, the ball experienced a maximum overshoot of 25%, and within 2 seconds, it settled at the desired distance of 5 cm.

After designing the controller for linearized system, to ensure the designed state feedback controller is applicable to the real nonlinear system, we employed Simulink based on (4). The block diagram of such a system is defined in Simulink as follows.

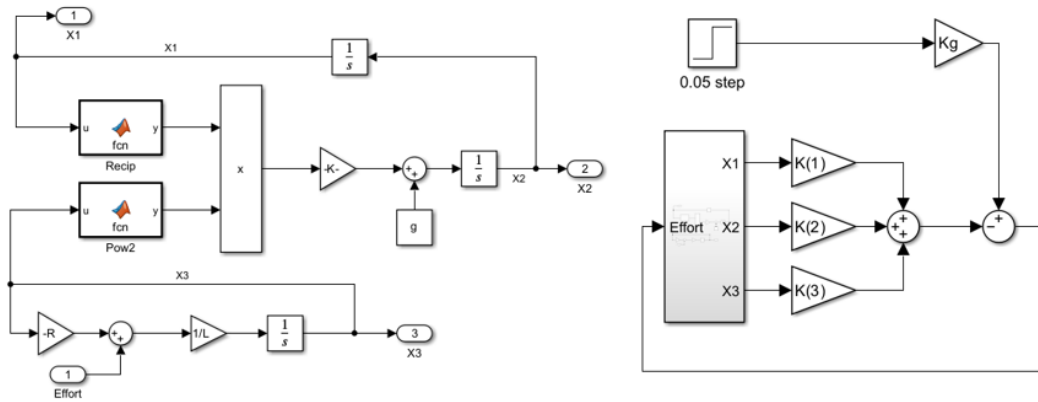


Figure 4 A scheme of the original nonlinear system defined in Simulink

In the figure above, MATLAB function blocks are provided to define the inversion and the signal raised to the power of two. Other blocks are defined according to the nonlinear dynamics of the system. Additionally, the state feedback block is defined as shown in the following figure.

The steady-state tracking error in the system arises because the designed controller lacks integral action, which is essential for completely eliminating errors in response to constant reference inputs or disturbances.

The feedforward gain K_g provides a correction for the desired reference input, but it does not account for the nonlinearities present in the system dynamics, such as the $\frac{x_3^2}{x_1}$ term. These nonlinearities introduce a mismatch between the expected and actual steady-state values, resulting in an offset. Furthermore, without

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an integral term in the control law, the system cannot dynamically compensate for this mismatch by accumulating the error over time, leaving the system unable to drive the steady-state error to zero.

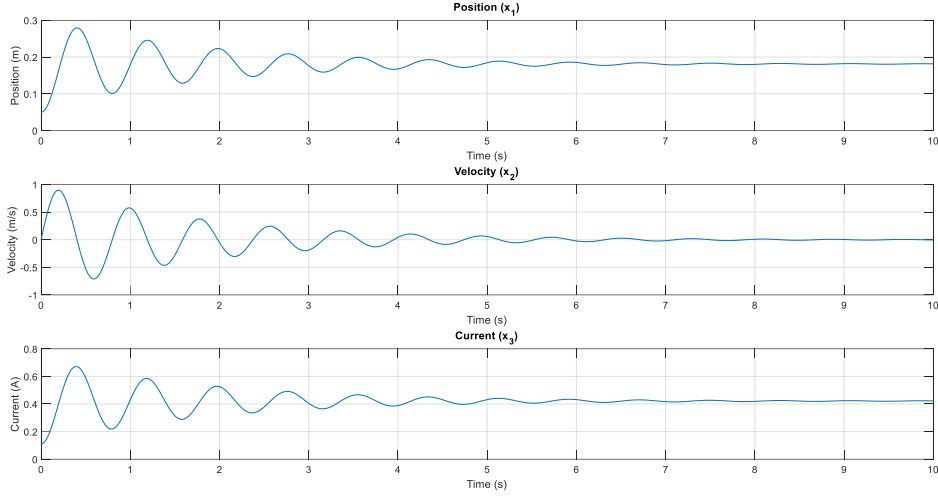


Figure 5 Results achieved after employing the designed controller on the actual nonlinear system.

Although the velocity eventually converges to zero, its fluctuations make the overall nonlinear dynamics unstable. One possible approach to make the system more robust is to incorporate an integrator term into the state feedback controller. The controller effort structure is described in (15).

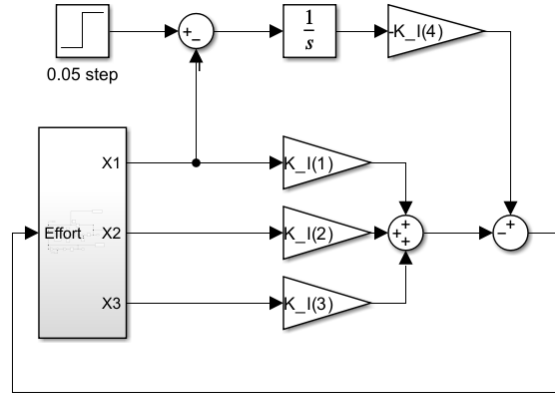


Figure 6 State feedback and integral controller

$$u = -Kx + K_I x_r; \dot{X}_r = r - y$$

$$A_{cl} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} - \begin{bmatrix} B \\ 0 \end{bmatrix} [K \quad K_I]$$

$$A_l = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}; B_l = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

(15)

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The controllability matrix for such a system in this case has rank of 4, which is corresponding to a full rank observability matrix and the states are fully observable. Then, we can implement the integral controller. Then, we can implement the integral controller. Using a similar approach as in the previous section, we now select a fourth pole far from the imaginary axis to reduce the transient effects of the pole. As a result, we obtain the following K matrix. $K_I = [-0.3762 \quad -0.0333 \quad -9.8500 \quad 0.0831]$

Which means that $K_I = -0.0831$.

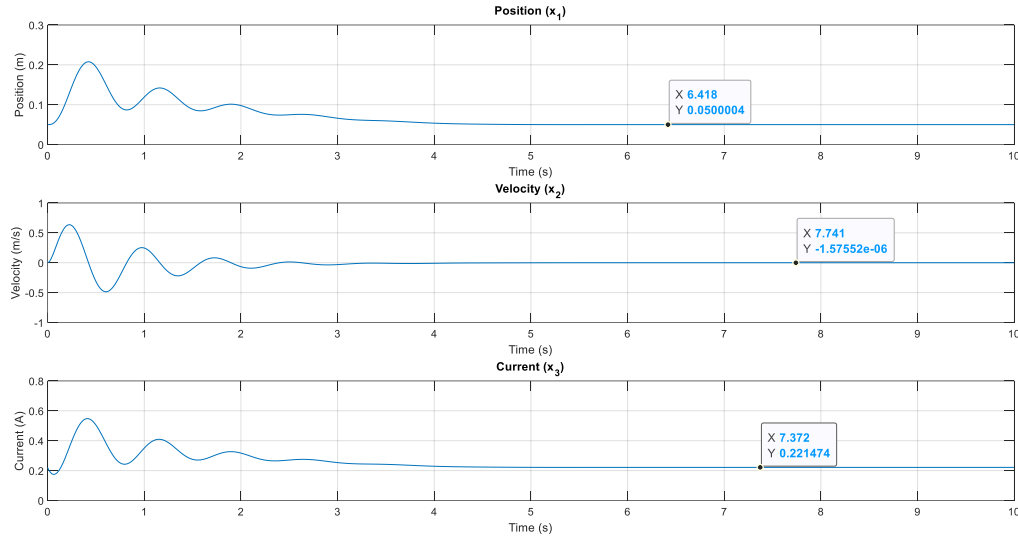


Figure 7 Improved tracking performance of the nonlinear system after incorporating the integrator controller

This time, we observe that although there is a transient at the beginning of the simulation, which arises from the nonlinear nature of the problem, the integral controller successfully accounts for the steady-state error observed in the previous part. This time, we aim to design a state-feedback controller, assuming the states are estimated using an appropriate observer. First, we verify the observability of the linearized system. To do this, we construct the observability matrix to determine if the system states can be accurately estimated from the output.

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 196.2 & 0 & -88.59 \end{bmatrix}$$

The observability matrix is full-rank; therefore, the states are observable. The controllability of the states was examined previously. Moreover, due to the limitations observed in the previous section, we aim to define an observer-based integral control. Considering the dynamics of the linearized model.

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$$\dot{x} = Ax + Bu; y = Cx$$

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly; \hat{y} = C\hat{x}$$

$$u = -K\hat{x} + K_I x_r \rightarrow$$

$$\begin{bmatrix} \dot{x} \\ \dot{x}_r \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & BK_I & -BK \\ -C & 0 & 0 \\ LC & BK_I & A - BK - LC \end{bmatrix} \begin{bmatrix} x \\ x_r \\ \hat{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} r, y = \begin{bmatrix} C & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ x_r \\ \hat{x} \end{bmatrix}$$

(16)

The block diagram representing this dynamic system is shown below.

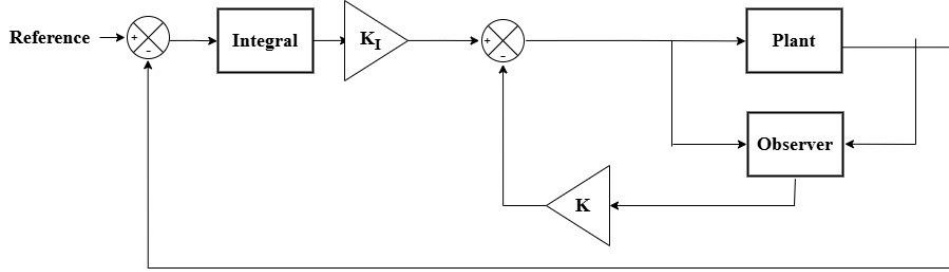


Figure 8 Block diagram of observer based integral control

The important point here is that the state feedback and integral controller are designed in an independent procedure compared to the observer design. The state feedback and integral controllers were previously defined, so we only need to define an appropriate observer.

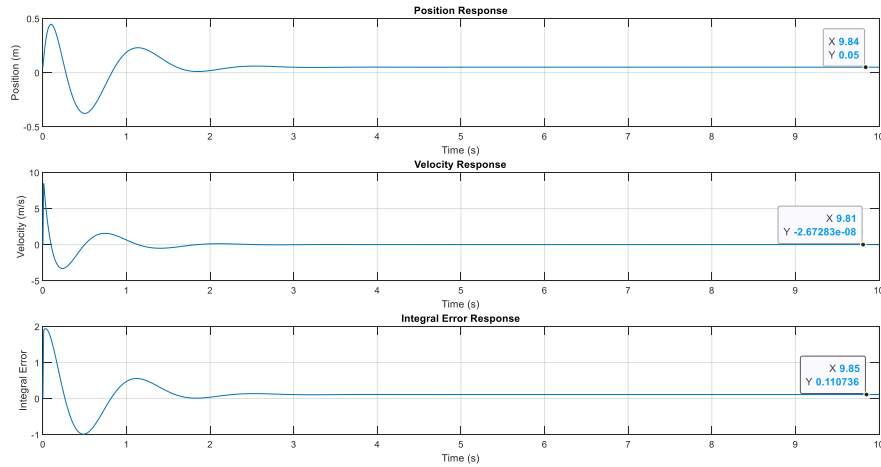


Figure 9 States performance after implementing observer

In this case, the observer gains are considered to be very large, as the states need to be estimated quickly and then implemented in the feedback controller design. In this configuration, the control policy uses the estimated state instead of the actual states, and the following results are achieved. Clearly, implementing the observer leads to poor transient performance in the initial time steps. This is due to the weakness of the

observer in estimating the actual states at the beginning. On the other hand, in cases where the states are not available, the designed observer is a means of estimating the states for the implementation of various control policies.

5- Conclusion

This project involved the modeling, analysis, and control of a single-axis magnetic levitation system, demonstrating the design and implementation of some control strategies to achieve system stability and performance. Key design decisions included the linearization of the inherently nonlinear system dynamics around an equilibrium point, enabling the use of state-space representation and facilitating control methods like state feedback and integral control. These approaches were employed for meeting specific performance goals, such as reducing steady-state errors. One of the primary challenges encountered was the system's inherent instability, which required selection of control parameters and iterative testing to achieve desired outcomes. Modeling inaccuracies and the nonlinear nature of the system also introduced complexities, particularly during the transition from simulation to the real nonlinear model. The incorporation of an observer to estimate unmeasurable states can be a valuable approach for cases where accessibility to states is not plausible. We gained a deeper understanding of the interplay between theory and application, particularly in implementing state feedback and observer-based control strategies. Additionally, the experience highlighted the importance of advanced tools like MATLAB/Simulink for simulating complex dynamic systems. A major takeaway was the significance of robust control designs that balance transient performance and steady-state accuracy. Future work on this project could involve exploring more advanced nonlinear control methods, such as adaptive or sliding mode control, to enhance robustness against disturbances and model uncertainties. Extending the system to multi-axis magnetic levitation would offer an opportunity to apply these control principles in a more complex context. Additionally, addressing real-world challenges, such as sensor noise and external disturbances, would further refine the system's practical applicability.

6- References

- [1] Hyung-Woo Lee, Ki-Chan Kim, and Ju Lee, "Review of maglev train technologies," *IEEE Trans. Magn.*, vol. 42, no. 7, pp. 1917–1925, Jul. 2006, doi: 10.1109/TMAG.2006.875842.
- [2] E. Beyazit and M. Givoni, "High Speed Rail, Technology Development of," in *Encyclopedia of Sustainability Science and Technology*, New York, NY: Springer New York, 2012, pp. 4929–4945. doi: 10.1007/978-1-4419-0851-3_902.
- [3] A. Abbas *et al.*, "Design and Control of Magnetic Levitation System," in *2019 International Conference on Electrical, Communication, and Computer Engineering (ICECCE)*, IEEE, Jul. 2019, pp. 1–5. doi: 10.1109/ICECCE47252.2019.8940711.
- [4] Y. Sun, J. Xu, C. Chen, and W. Hu, "Reinforcement Learning-Based Optimal Tracking Control for Levitation System of Maglev Vehicle With Input Time Delay," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022, doi: 10.1109/TIM.2022.3142059.
- [5] M. Spielmann, P. de Haan, and R. W. Scholz, "Environmental rebound effects of high-speed transport technologies: a case study of climate change rebound effects of a future underground maglev train system," *J. Clean. Prod.*, vol. 16, no. 13, pp. 1388–1398, Sep. 2008, doi: 10.1016/j.jclepro.2007.08.001.
- [6] Faa-Jeng Lin, Syuan-Yi Chen, and Kuo-Kai Shyu, "Robust Dynamic Sliding-Mode Control Using Adaptive RENN for Magnetic Levitation System," *IEEE Trans. Neural Networks*, vol. 20, no. 6, pp. 938–951, Jun. 2009, doi: 10.1109/TNN.2009.2014228.
- [7] Z. Qadir, A. Munir, T. Ashfaq, H. S. Munawar, M. A. Khan, and K. Le, "A prototype of an energy-efficient MAGLEV train: A step towards cleaner train transport," *Clean. Eng. Technol.*, vol. 4, p. 100217, Oct. 2021, doi: 10.1016/j.clet.2021.100217.
- [8] Rong-Jong Wai, Jing-Xiang Yao, and Jeng-Dao Lee, "Backstepping Fuzzy-Neural-Network Control Design for Hybrid Maglev Transportation System," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 26, no. 2, pp. 302–317, Feb. 2015, doi: 10.1109/TNNLS.2014.2314718.
- [9] Z. Zhang and X. Li, "Real-Time Adaptive Control of a Magnetic Levitation System with a Large Range of Load Disturbance," *Sensors*, vol. 18, no. 5, p. 1512, May 2018, doi: 10.3390/s18051512.
- [10] F. B. J. Gilson and A. L. B. José, "PID Control Design for a Maglev Train System," *Appl. Mech. Mater.*, vol. 389, pp. 425–429, Aug. 2013, doi: 10.4028/www.scientific.net/AMM.389.425.
- [11] S. Verma, W. Kim, and H. Shakir, "Multi-Axis Maglev Nanopositioner for Precision Manufacturing and Manipulation Applications," *IEEE Trans. Ind. Appl.*, vol. 41, no. 5, pp. 1159–1167, Sep. 2005, doi: 10.1109/TIA.2005.853374.
- [12] E. C. Ozkat, M. Abdioglu, and U. K. Ozturk, "Machine learning driven optimization and parameter selection of multi-surface HTS Maglev," *Phys. C Supercond. its Appl.*, vol. 616, p. 1354430, Jan. 2024, doi: 10.1016/j.physc.2023.1354430.

Name	Contribution
A.Mashhadireza	State feedback controller desing-Observer desing-Simulink and simulations-Final project edition
R.Suhail	Lyapunov stability Analysis-Output feedback controller desing-Dynamic Modeling
H. Ali Seyes	Dynamic Modeling-Linearization
M. Polagoni	Introduction